B. TECH-ME-III-II Sem

23A03601P	HEAT TRANSFER LAB	L	T	P	C
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Course Objectives: Students undergoing this course would	
1	Understand different modes of heat transfer
2	Gain knowledge about natural and force convection phenomenon
3	Estimate experimental uncertainty in measurements

Course Outcomes: Upon the successful completion of course, students will be able to	
1	Explain different modes of heat transfer
2	Identify parameters for measurement for calculating heat transfer
3	Determine effectiveness of heat exchanger
4	Design new equipment related to heat transfer
5	Apply principles of heat transfer in wide application in industries.

List of Experiments:

1. Determine the overall heat transfer coefficient across the width of composite wall
2. Determine the thermal conductivity of a metal rod
3. Determine the thermal conductivity of insulating powder material through concentric sphere apparatus
4. Determine the thermal conductivity of insulating material through lagged pipe apparatus
5. Determine the efficiency of a pin fin in natural and forced convection.
6. Determine the heat transfer coefficient for a vertical cylinder in natural convection
7. Determine the heat transfer coefficient in forced convection of air in a horizontal tube.
8. Determine the heat transfer coefficients on film and drop wise condensation apparatus.
9. Determine the effectiveness of a parallel and counter flow heat exchanger.
10. Study the pool boiling phenomenon and different regimes of pool boiling.
11. Experiment on pool boiling
12. Determine the emissivity of the test plate surface.
13. Experiment on Stefan-Boltzmann apparatus

14. Determine the heat transfer rate coefficient in fluidized bed apparatus.

Virtual Lab:-

1. Determination of thermal conductivity of a metal rod

   https://sites.google.com/view/vlab-bnmitmech/home/heat-transfer-lab/determination-of-thermal-conductivity-of-a-metal-rod
2. Natural Convection heat transfer

   https://sites.google.com/view/vlab-bnmitmech/home/heat-transfer-lab/natural-convection
3. Heat Transfer by Radiation

   https://vlab.amrita.edu/index.php?sub=1&brch=194&sim=802&cnt=1
4. Heat transfer by Conduction

   https://vlab.amrita.edu/index.php?sub=1&brch=194&sim=801&cnt=1
5. The Study of phase change

   https://vlab.amrita.edu/index.php?sub=1&brch=194&sim=709&cnt=1
6. Black Body Radiation: Determination of Stefan's Constant

   https://vlab.amrita.edu/index.php?sub=1&brch=194&sim=548&cnt=1
7. Newton's Law of Cooling

   https://vlab.amrita.edu/index.php?sub=1&brch=194&sim=354&cnt=1
8. Lee's Disc Apparatus

   https://vlab.amrita.edu/index.php?sub=1&brch=194&sim=353&cnt=1
9. Thermo Couple-See beck Effect
10. https://vlab.amrita.edu/index.php?sub=1&brch=194&sim=351&cnt=1